

ALGEBRAIC STRUCTURE
Group definition and examples

Lecture-1

ALGEBRAIC STRUCTURE



- binary operation & algebraic structure

➤ A **binary operation** on a set combines two elements of the set to produce another element of the set.

$$a * b \in G, \forall a, b \in G$$

e.g. $+$, $-$, $\cdot$, $/$ are binary operations

➤ The combination of the set and the operations that are applied to the elements of the set is called an **algebraic structure**.

➤ e.g. $(\mathbb{N}, +)$, $(\mathbb{Z}, -)$, $(\mathbb{R}, +, \cdot)$ are the algebraic structures

- Properties on Operations

- **Closure Property**

$$a * b \in G, \forall a, b \in G$$

e.g. $\mathbb{N}$ is closed w.r.t. addition & multiplication

- **Associative Property**

$$(a * b) * c = a * (b * c), \forall a, b, c \in G$$

- Properties on Operations

➤ **Commutative Property**

$$a * b = b * a, \forall a, b \in G$$

e.g. The set of integers $\mathbb{Z}$ is associative and commutative w.r.t. addition & multiplication

➤ **Existence of Identity** $\exists e \in G$ such that

$a * e = e * a = a, \forall a \in G$, then e is called identity element of G w.r.t. operation $*$

e.g. 0 is the additive identity

1 is the multiplicative identity

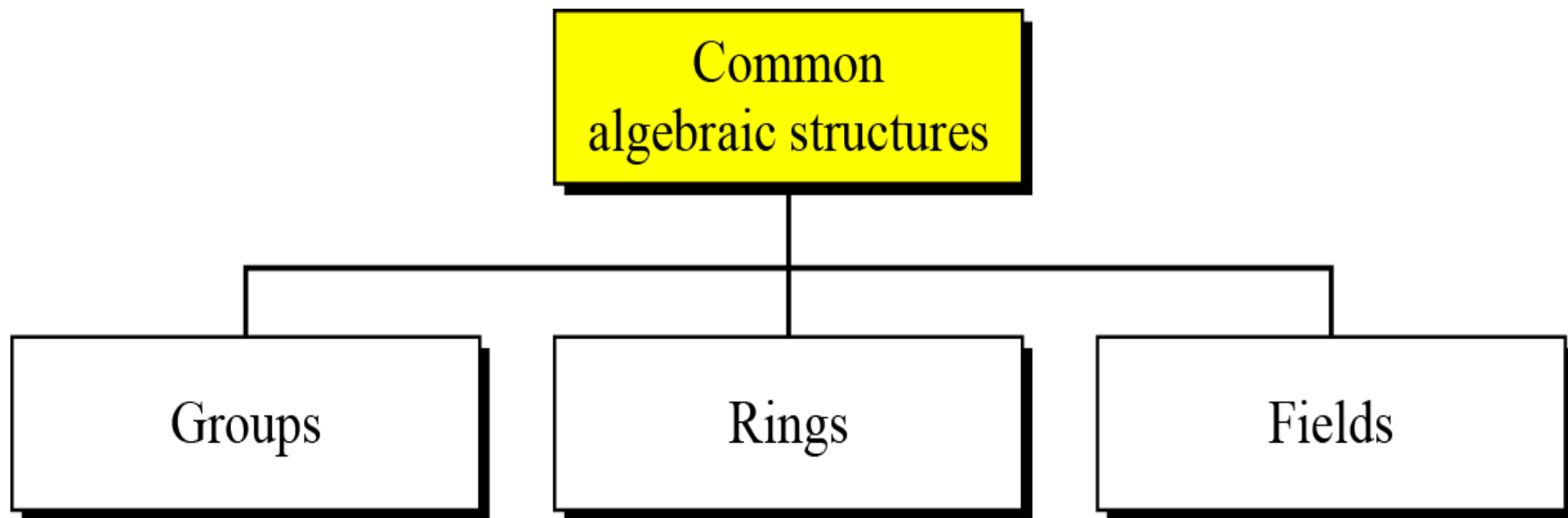
- Properties on Operations

➤ **Existence of Inverse** $\forall a \in G, \exists b \in G$ such that $a * b = e = b * a$, then a and b is the inverses of each other under operation $*$

e.g. -3 and 3 are the additive inverses

3 and $\frac{1}{3}$ are the multiplicative inverses

- Group



• Semi- Group & Monoid

Groupoid: Let S be a non-empty set with operation $*$. Then $(S, *)$ is called Groupoid if it satisfies –
Closure Property

Semi-Group

Let S be a non-empty set with operation $*$. Then $(S, *)$ is called semi-group if it satisfies –
Closure Property
Associative Property

Monoid

Let S be a non-empty set with operation $*$. Then $(S, *)$ is called monoid if it satisfies –
Closure Property
Associative Property
Existence of Identity Element

- Group & abelian or commutative group

Let G be a non-empty set with operation $*$. Then $(G, *)$ is called group if it satisfies the properties –

- Closure
- Associative
- Existence of Identity Element
- Existence of Inverse

A group is called **abelian or commutative group** if it satisfies the commutative property.

- Group & abelian group : examples

- ❖ The set $\mathbb{Z}^+$ under addition is a group.
- ❖ The set of all nonnegative integers under addition is not a group.
- ❖ The set $\mathbb{Z}^+$ under multiplication is not a group.

The set $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ under addition are abelian groups.

- Group & abelian group : examples

- ❖ The set $\mathbb{Z}^+$ under addition is not a group.
- ❖ The set of all nonnegative integers under addition is not a group.
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The set $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ under addition are abelian groups.

- Group & abelian group : examples

- ❖ $G = (1, \omega, \omega^2)$ is an abelian group w.r.t. multiplication.
- ❖ $G = (1, -1, i, -i)$ w.r.t. multiplication is abelian.

- Finite , Infinite Group & order of the Group

- A group $(G,*)$ is called **finite** if G consists of a finite number of distinct elements.

Ex. $G=\{1,-1,i,-i\}$ w.r.t. $\times$

- A group is called **infinite** if it is not finite.

Ex. $(\mathbb{Z},+), (\mathbb{R},.)$

- The number of elements in a group is called **order** of the group. $o(G) = 4, o(\mathbb{Z}) = \infty$

Theorem : The identity element in a group is unique.

Proof : Suppose e and e' are two identity elements of a group $(G, *)$

$\therefore e$ and e' are the elements of G

If e is the identity element, then

$$e * e' = e' \quad \dots\dots(1)$$

If e' is the identity element, then

$$e * e' = e \quad \dots\dots(2)$$

From eq. (1) and (2),

$$e = e'$$

Hence the identity element of a group is unique.

Properties of group.

- The identity element is its own inverse $e^{-1} = e$
- The inverse of each element of a group is unique.
- If the inverse of a is a^{-1} , then the inverse of a^{-1} is a
i.e., $(a^{-1})^{-1} = a$
- Let $(G, *)$ be a group. Prove that $(a*b)^{-1} = b^{-1}*a^{-1}$, $\forall a, b \in G$
- If a, b, c are the elements of a group G , then
$$a * b = a * c \Rightarrow b = c \quad (\text{Left Cancellation law})$$
$$b * a = c * a \Rightarrow b = c \quad (\text{Right Cancellation law})$$
- if $a^2 = a$, $a \in G$, then $a = e$.

Properties of group.

- ❖ if for every element a in a group G , $a^2 = e$, then G is an abelian group.
- ❖ if every element of a group G is its own inverse, then G is abelian.
- ❖ if a, b are any two elements of a group G , then $(a*b)^2 = a^2*b^2$ iff G is abelian.
- ❖ a group G is abelian if
$$b^{-1}*a^{-1}*b*a = e, \quad \forall a, b \in G$$

Theorem : The inverse of each element of a group is unique.

Proof : Let $(G, *)$ be a group and e be the identity of G

Let $a \in G$

Suppose b and c are two inverses of a

$$\therefore b * a = e = a * b$$

$$\text{and } c * a = e = a * c$$

$$\begin{aligned} b * (a * c) &= b * e && [\square \ a * c = e] \\ &= b && [\square \ e \text{ is the identity}] \end{aligned}$$

$$\begin{aligned} (b * a) * c &= e * c && [\square \ b * a = e] \\ &= c && [\square \ e \text{ is identity}] \end{aligned}$$

In a group, composition $*$ is associative.

$$\therefore b * (a * c) = (b * a) * c$$

$$\text{Hence } b = c$$

Theorem : If the inverse of a is a^{-1} , then the inverse of a^{-1} is a i.e., $(a^{-1})^{-1} = a$

Proof: If e is the identity element, we have

$$a^{-1} * a = e \quad [\text{by definition of inverse}]$$

Multiplying both sides on the left by $(a^{-1})^{-1}$ which is an element of G because a^{-1} is an element of G , we get

$$(a^{-1})^{-1} * (a^{-1} * a) = (a^{-1})^{-1} * e$$

$$((a^{-1})^{-1} * a^{-1}) * a = (a^{-1})^{-1}$$

$[* \text{ is associative \& } e \text{ is identity element}]$

$$e * a = (a^{-1})^{-1}$$

$[(a^{-1})^{-1} \text{ is inverse of } a^{-1}]$

$$a = (a^{-1})^{-1}$$

$$(a^{-1})^{-1} = a$$

Theorem : Let $(G, *)$ be a group. Prove that $(a*b)^{-1} = b^{-1}*a^{-1}$, $\forall a, b \in G$

Proof : Let e be the identity element of a group G .

Suppose a and b are elements of G . If a^{-1} and b^{-1} are the inverses of a and b respectively, then

$$a^{-1} * a = e = a * a^{-1}$$

$$b^{-1} * b = e = b * b^{-1}$$

Now, $(a*b) * (b^{-1}*a^{-1})$

$$= ((a * b) * b^{-1}) * a^{-1} \quad [\text{By Associativity}]$$

$$= (a * (b * b^{-1})) * a^{-1} \quad [\text{By Associativity}]$$

$$= (a * e) * a^{-1} \quad [\square \ b * b^{-1} = e]$$

$$= a * a^{-1} \quad [\square \ a * e = a]$$

$$= e \quad [\square \ a * a^{-1} = e]$$

Also, $(b^{-1}*a^{-1}) * (a * b) = b^{-1} * [a^{-1} * (a * b)]$ [by associativity]

$$= b^{-1} * [(a^{-1}*a)*b] = b^{-1} * [e*b] = b^{-1} * b = e$$

Thus, $(b^{-1}*a^{-1})*(a*b) = e = (a*b)*(b^{-1}*a^{-1})$

So, by definition of inverse , we have $(a*b)^{-1} = b^{-1}*a^{-1}$

Theorem : If a, b, c are the elements of a group G , then

$$a * b = a * c \Rightarrow b = c \quad (\text{Left Cancellation law})$$

$$b * a = c * a \Rightarrow b = c \quad (\text{Right Cancellation law})$$

Proof : $a \in G \Rightarrow \exists a^{-1} \in G$ such that $a^{-1} * a = e = a * a^{-1}$ where e is the identity element.

$$\text{Now, } a * b = a * c$$

Multiplying both sides on left by a^{-1} , we get

$$a^{-1} * (a * b) = a^{-1} * (a * c)$$

$$\Rightarrow (a^{-1} * a) * b = (a^{-1} * a) * c \quad [\text{by associativity}]$$

$$\Rightarrow e * b = e * c \quad [\square a^{-1} * a = e]$$

$$\Rightarrow b = c \quad [\square e \text{ is the identity}]$$

$$\text{Also, } b * a = c * a$$

$$\Rightarrow (b * a) * a^{-1} = (c * a) * a^{-1}$$

$$\Rightarrow b * (a * a^{-1}) = c * (a * a^{-1})$$

$$\Rightarrow b * e = c * e \Rightarrow b = c$$

Theorem : If a, b are any two elements of a group G , then the equations $a * x = b$ and $y * a = b$ have unique solutions in G .

Proof : $a \in G \Rightarrow \exists a^{-1} \in G$

$a^{-1} \in G, b \in G \Rightarrow a^{-1} * b \in G$ [by closure property] Substituting $a^{-1} * b$ for x in the equation $a * x = b$, we have

$$a * (a^{-1} * b) = (a * a^{-1}) * b = e * b = b.$$

Thus $x = a^{-1} * b$ is a solution in G of the equation $a * x = b$.

To show that the solution is unique, let us suppose that

x_1 and x_2 are two solutions of the equation $a * x = b$.

Then,

$$a * x_1 = b \text{ and } a * x_2 = b$$

$$\Rightarrow a * x_1 = a * x_2$$

$$\Rightarrow x_1 = x_2 \quad [\text{By left cancellation law}]$$

Therefore, the solution is unique.

To prove that the equation $y * a = b$ has a unique solution in G

Ex : Prove that if $a^2 = a$, $a \in G$, then $a = e$.

Sol: We have $a^2 = a$
 $\Rightarrow a * a = a$
 $\Rightarrow a * a = a * e$
 $\Rightarrow a = e$

Ex : Given $axa = b$ in G , find x .

Sol: We have $axa = b$
 $\Rightarrow a^{-1}(axa) = a^{-1}b$
 $\Rightarrow (a^{-1}a)(xa) = a^{-1}b$
 $\Rightarrow e(xa) = a^{-1}b \Rightarrow xa = a^{-1}b$
 $\Rightarrow (xa)a^{-1} = (a^{-1}b)a^{-1} \Rightarrow x(aa^{-1}) = a^{-1}ba^{-1}$
 $\Rightarrow xe = a^{-1}ba^{-1}$
 $\Rightarrow x = a^{-1}ba^{-1}$

Ex : Prove that if for every element a in a group G , $a^2 = e$, then G is an abelian group.

Sol : Let $a, b \in G \Rightarrow ab \in G$.

Therefore, $(ab)^2 = e$

$$\square (ab)^2 = e$$

$$\Rightarrow (ab)^*(ab) = e$$

$$\Rightarrow (ab)^{-1} = ab$$

$$\Rightarrow b^{-1} a^{-1} = ab. \dots\dots\dots(1)$$

$$\square a^2 = e \Rightarrow a^*a = e \Rightarrow a^{-1} = a$$

$$\text{Similarly, } b^2 = e \Rightarrow b^{-1} = b$$

From eq. (1),

$$b^*a = a^*b.$$

Thus, $a^*b = b^*a \quad \forall a, b \in G$

Hence G is an abelian group

Ex. : Show that if every element of a group G is its own inverse, then G is abelian.

Sol : Let $a, b \in G \Rightarrow a * b \in G$

Given that every element of G is its own inverse

$$\Rightarrow (a*b)^{-1} = a*b$$

$$\Rightarrow b^{-1}*a^{-1} = a*b$$

$$\Rightarrow b*a = a*b \quad [\square \ a^{-1} = a, \ b^{-1} = b]$$

Thus, we have $a*b = b*a \ \forall \ a, b \in G$.

Therefore, G is an abelian group.

Ex. : Show that if a, b are any two elements of a group G , then $(a*b)^2 = a^2*b^2$ iff G is abelian.

Sol : Suppose G is an abelian group

$$\begin{aligned}(a*b)^2 &= (a*b)*(a*b) = a*(b*a)*b \\ &= a*(a*b)*b \quad [\square \text{ by commutative property}] \\ &= (a*a)*(b*b) = a^2*b^2\end{aligned}$$

Converse: Suppose $(a*b)^2 = a^2 * b^2$

$$\Rightarrow (a*b) * (a*b) = (a*a) * (b*b)$$

$$\Rightarrow a * (b*a) * b = a * (a*b) * b \quad [\text{by associativity}]$$

$$\Rightarrow (b*a)*b = (a*b)*b \quad [\text{by left cancellation law}]$$

$$\Rightarrow b*a = a*b \quad [\text{by right cancellation law}]$$

$$\Rightarrow G \text{ is abelian}$$

Ex. Prove that a group G is abelian if

$$b^{-1} * a^{-1} * b * a = e, \quad \forall a, b \in G$$

Sol : We have

$$b^{-1} * a^{-1} * b * a = e$$

$$\Rightarrow (b^{-1} * a^{-1}) * (b * a) = e$$

$$\Rightarrow (b^{-1} * a^{-1})^{-1} = b * a$$

$$\Rightarrow (a^{-1})^{-1} * (b^{-1})^{-1} = b * a$$

$$\Rightarrow a * b = b * a$$

$$\Rightarrow G \text{ is abelian.}$$

$$[\square \ a * b = e \Rightarrow a^{-1} = b]$$

$$[\square \ (a * b)^{-1} = b^{-1} * a^{-1}]$$

$$[\square \ (a^{-1})^{-1} = a]$$

Ex. : A group of order 4 is abelian.

Sol :

Let $G = \{e, a, b, c\}$ be a group of order four where e is the identity element $\Rightarrow e^{-1} = e$

There must be at least one more element in G which is its own inverse. Let $a^{-1} = a$.

Case -1: If $b^{-1} = b$ and $c^{-1} = c$, then G is abelian.

Case - 2: If $b^{-1} = c$, then $c^{-1} = b$, then

$$b*c = e = c*b.$$

Also, $a^{-1} = a \Rightarrow a*a = e$.

In this case,

composition table for G will be as follows:

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

$a*b$ can not be equal to a or b .

if $a*b = b \Rightarrow a = e$ which is not possible

$$\therefore a*b = c$$

Similarly, $a*c = b$

From the table, it can be seen that composition in G is commutative.

Therefore G is abelian.